

Assignment-2

Last Date of Submission: 08-12-2025

- 1 A subnet uses mask 255.255.255.192 (/26). Compute the number of subnets, hosts per subnet, and broadcast address for subnet having network ID 10.10.4.64.
- 2 A network with address 172.16.0.0/16 must be divided into 500 subnets. Determine the new prefix length and maximum hosts per subnet.
- 3 In distance vector routing, node A receives the following distances from its neighbours:

Destination From B From C

D	4	7
E	2	5

If the cost of link A–B is 3 and A–C is 2, compute A's updated distance to D and distance to E.

- 4 Given a routing table:

Destination Mask Interface

10.0.0.0	/8	eth0
10.1.0.0	/16	eth1
10.1.2.0	/24	eth2

Determine the output interface for destination 10.1.2.55 and 10.2.8.7.

- 5 A packet experiences the following delays in a network:

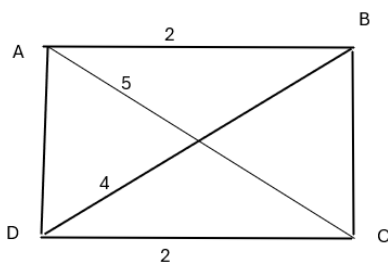
Processing delay: 3 μ s
 Queuing delay: 2.5 ms
 Transmission delay: 0.4 ms
 Propagation delay: 10 ms

Compute the total end-to-end delay.

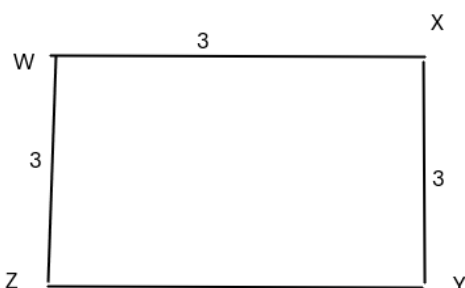
- 6 Given network 192.168.0.0/24, create 6 subnets of equal size. Give the subnet masks, network IDs, usable host ranges, and broadcast addresses.

- 7 For network 10.0.0.0/8, allocate subnets to satisfy host requirements: 1000 hosts, 400 hosts, 60 hosts, and 10 hosts. Provide subnet prefixes, network IDs, and usable host ranges for each allocation using VLSM.

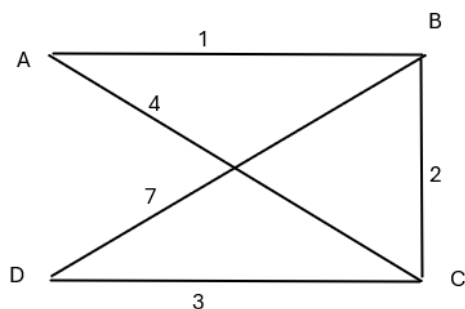
Distance Vector: Nodes A, B, C, D; link costs: A–B=2, A–C=5, B–C=1, B–D=4, C–D=2. Starting from initial direct distances, compute A's distance vector to all nodes after one full exchange with neighbors B and C. Show intermediate sums used.



- 8 Distance Vector Convergence: Given a ring of 4 nodes (W–X–Y–Z–W) with equal link cost 3, what is the distance from W to Y after DV converges? Show steps of path cost addition.



- 9 Link State: For a network with nodes {A,B,C,D} and link weights A-B=1, A-C=4, B-C=2, B-D=7, C-D=3, run Dijkstra from A and list the final shortest path cost and predecessor for each node. (Show numeric costs and chosen predecessors.)



- 10 Forwarding table longest-prefix: Given forwarding entries:
 10.0.0.0/8 → iface0
 10.1.0.0/16 → iface1
 10.1.2.0/24 → iface2
 0.0.0.0/0 → gw1
 For destination 10.1.2.129 and 10.1.3.5, specify chosen interface/gateway and explain (by prefix length comparison) why.
- 11 A TCP connection uses MSS = 1200 bytes and header size = 40 bytes (TCP+IP). If the application sends 18,000 bytes of data, determine how many segments are created and the size of each transmitted packet.
- 12 A TCP sender has cwnd = 12 KB, rwnd = 20 KB, MSS = 1000 bytes. Determine the effective window size and the maximum data (in bytes) the sender can transmit before waiting for ACKs.
- 13 A TCP connection starts with cwnd = 1 MSS. Threshold = 12 MSS. Show the value of cwnd after each RTT for the first 10 RTTs.
- 14 Sender sends 4 segments. ACKs arrive for segments 1 and 2, but segment 3 is lost. If three duplicate ACKs are received for segment 2, identify:
 1. Which segment number is retransmitted?
 2. What happens to cwnd and ssthresh numerically?
- 15 Explain the difference between connection-oriented and connectionless transport services, with examples.
- 16 Define multiplexing and demultiplexing and describe how ports are used in transport layer protocols.
- 17 Describe the three-way handshake in TCP and explain why a two-way handshake is insufficient.
- 18 Compare TCP and UDP in terms of header fields, reliability, and applications.
- 19 Explain the basic steps involved in establishing a TCP connection between a client and a server using sockets.
- 20 Write the purpose of the following socket functions: socket(), bind(), listen(), accept(), connect().
- 21 Name any three application-layer protocols and state their use.
- 22 Explain the difference between persistent and non-persistent HTTP connections. Include how many TCP connections are created in each mode.
- 23 What is compression? Give one example where presentation layer uses it.

